higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

MECHANOTECHNICS N6

(8190236)

21 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages, 2 diagram sheets and a formula sheet of 3 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MECHANOTECHNICS N6
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Questions can be answered in any order, but subsections of questions must be kept together.
5. Start each question on a NEW page.
6. Draw a line after each completed subsection.
7. ALL calculations should have at least THREE steps, the formula, substitution and answer.
8. Use $g = 9{,}81\ m/s^2$.
9. Write neatly and legibly.

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QUESTION 1: BRAKES

The band brake shown on DIAGRAM SHEET 1, FIGURE 1, has to absorb 5 kW when the drum with a diameter of 400 mm is turning at 450 r/min clockwise. The centre of the drum and the pivot point of the lever are horizontally in line and 300 mm apart. The coefficient of friction between the band and the drum is 0,25. One end of the bend which is fixed to the lever is assumed to be perpendicular to the lever.

Calculate the following:

1.1 The angle of contact between the band and the braking drum (5)

1.2 The braking torque (2)

1.3 The force required at the end of the lever to stop the turning drum (7)

[14]

QUESTION 2: FRICTION CLUTCH

A plate clutch has 4 discs on the driving shaft and 3 discs on the driven shaft. Therefore there are six pairs of contact surfaces. The outside diameter and inside diameter of a friction clutch are 200 mm and 100 mm respectively. The power transmitted by the engaged clutch is 20 kW at 1 000 r/min. The coefficient of friction is 0,4 and assume uniform pressure theory.

Calculate the following:

2.1 The applied force on each pair

2.2 The stiffness of each spring in N/mm if there are six springs on each disc and 0,8 mm has worn away on each contact surface. Assume the new axial force after wearing would be 632,136 N.

(2 × 6) **[12]**

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QUESTION 3: LINE SHAFTS

On DIAGRAM SHEET 2, FIGURE 2 a line shaft is shown which is supported by two bearings 1,2 metres apart. The shaft is driven by pinion A with a PCD of 150 mm and which is transmitting 50 kW at 900 r/min. The driven gear B with a PCD of 280 mm is mounted 700 mm from the left bearing and is meshed with driver pinion A vertically below it. The pressure angle of the gears is 20˚.

A flywheel with a weight of 810 N is mounted 350 mm from the left bearing. A pulley with an effective diameter of 400 mm is mounted with a belt that is vertical and parallel and overhangs the right-hand bearing by 200 mm. The coefficient of friction between the belt and the pulley is 0,4 and the angle of contact is 180°. Neglect the weight of the pulley.

Calculate the following:

3.1	The torque transmitted by pinion A	(2)
3.2	The torque transmitted by the driven gear B	(4)
3.3	The tangential force between the gears	(1)
3.4	The radial force between the gears	(1)
3.5	The belt tensions on the slack and tight side	(5)
3.6	The vertical reaction on the bearings	(4)
		[17]

QUESTION 4: FLYWHEEL

An electric motor drives a flywheel which provides energy for a punching machine. The mass of the flywheel is 500 kg and its radius of gyration is 300 mm. The flywheel is mounted on a hollow shaft with internal and external diameters of 30 mm and 50 mm respectively. The shaft material has a density of 7 800 kg/m^3 and the length of the hollow shaft is 400 mm.

Calculate the following:

4.1	The moment of inertia of the shaft and the flywheel	(7)
4.2	The kinetic energy stored in the flywheel when rotating at 250 r/min	(2)
4.3	The linear velocity of the flywheel, if the flywheel has a diameter of 1,2 m	(2)
		[11]

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QUESTION 5: BALANCING

Four masses, A, B, C and D are attached on a rotating shaft. The magnitude of A, B and C is 5 kg, 7 kg and 6 kg respectively. Masses A, B, C and D are mounted at radii of 300 mm, 250 mm, 350 mm and 400 mm respectively. The axial spacing between A, B, C and D are equal. (**HINT:** Distance AB = BC = CD)

Determine the following using a suitable scale:

5.1 The angles between A and B and between A and C (11)

5.2 The magnitude of the unknown mass D (6)

[17]

QUESTION 6: DYNAMICS

A locomotive has a mass on the driving wheels. The coefficient of friction between the wheels and the rail is 0,25. The mass of the train is 400 tons. The rolling resistance for the locomotive and the train is 45 N/ton and 40 N/ton respectively. The locomotive and the train accelerate up an incline of 1 in 50 from rest to 50 km/h.

Calculate the following:

6.1 The tractive effect of the locomotive (2)

6.2 The time taken to accelerate from rest to 50 km/h (9)

6.3 The average power developed (4)

[15]

QUESTION 7: KINEMATICS

A slider-crank mechanism is shown on DIAGRAM SHEET 2, FIGURE 3. Crank AB is at 30° and rotates clockwise at 150 r/min. The lengths of the crank and links are as follows: AB = 100 mm, BC = 150 mm, DE = 120 mm and EF = 80 mm.

7.1 Draw the space diagram, using scale 25 mm = 1 cm. (4)

7.2 Calculate the velocity of crank AB at B. (2)

7.3 Draw the velocity diagram using scale 1 m/s = 40 mm. (6)

7.4 Calculate the line velocity of sliding block C. (2)

[14]

TOTAL: 100

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DIAGRAM SHEET 1

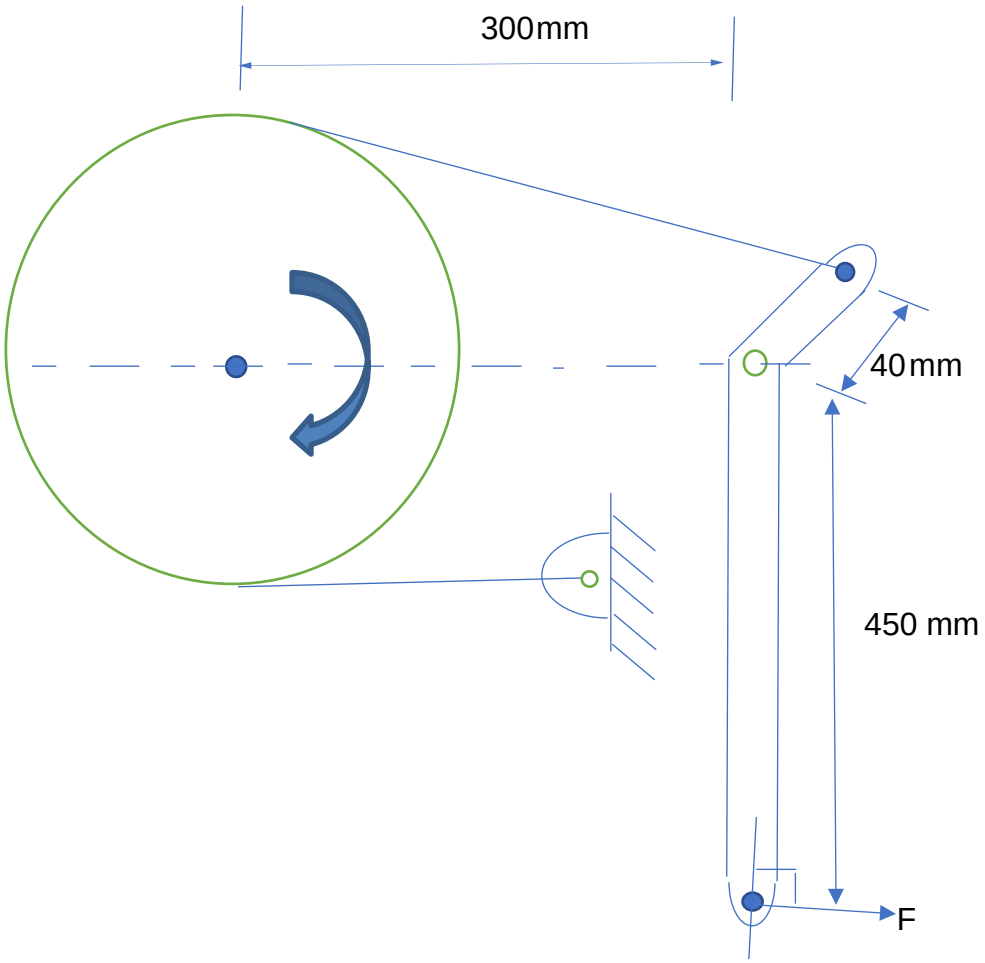


FIGURE 1

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DIAGRAM SHEET 2

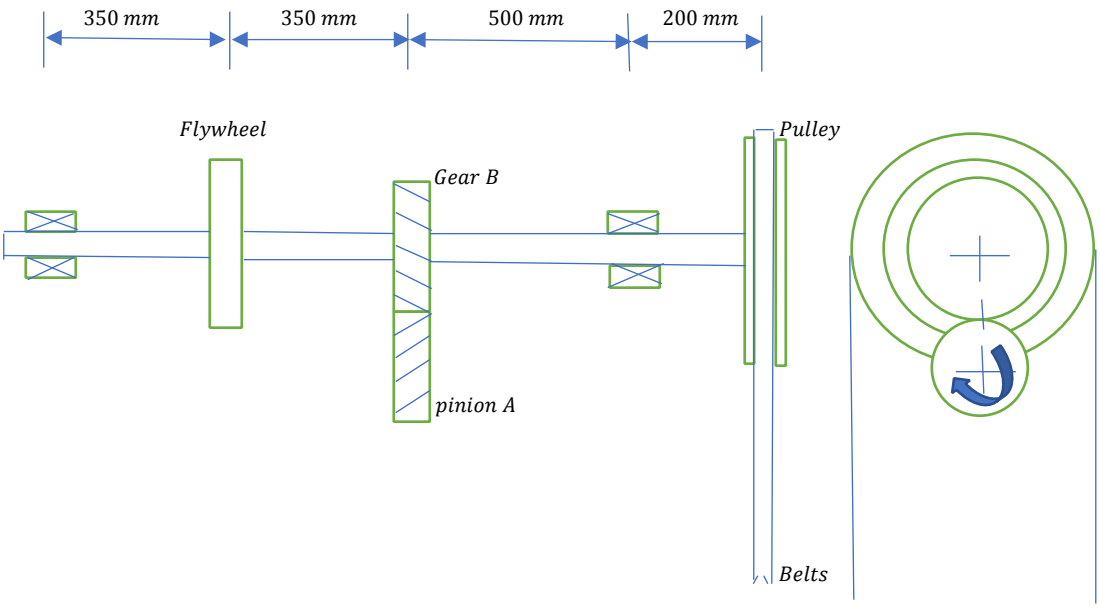


FIGURE 2

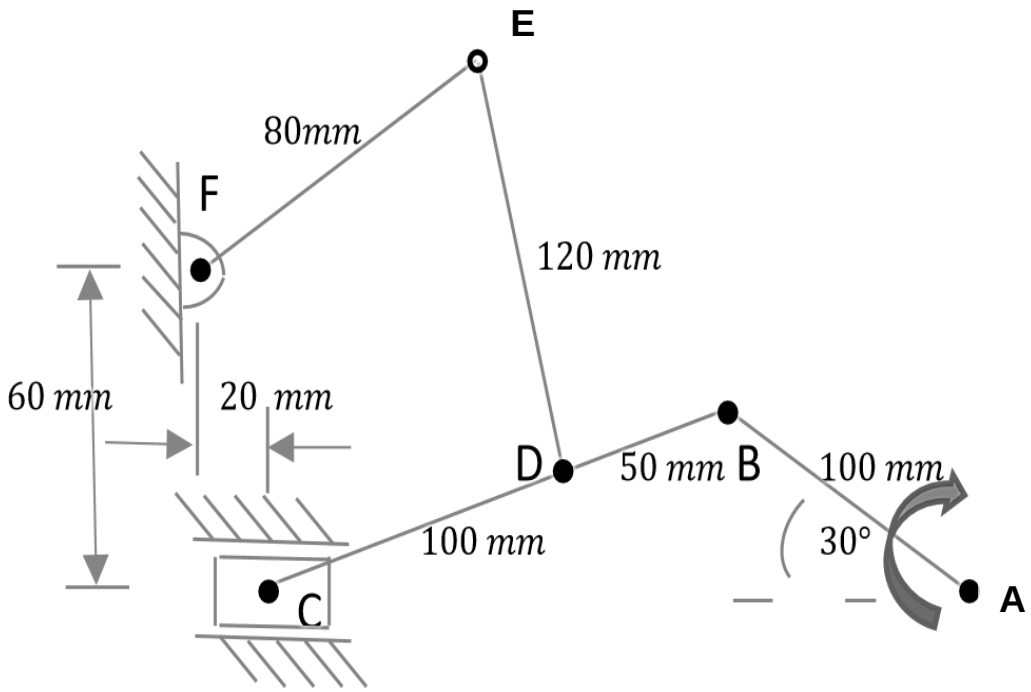


FIGURE 3

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MECHANOTECHNICS N6

FORMULA SHEET

1. $m = \frac{PCD}{T}$
2. $DO = m \times (T + 2)$
3. $C = \frac{m}{2} \times (TA + TB)$
4. $Ke = \frac{1}{2}mv^2$
5. $VR = \frac{TA}{TB}$
6. $VR = \frac{PCD\ of\ gear}{PCD\ of\ pinion}$
7. $VR = \frac{NB}{NA}$
8. $NA \times TA = NB \times TB$
9. $Ft = \frac{2 \times T}{PCD}$
10. $Fr = Ft \times \tan \emptyset$
11. $Fn = Ft \times \sec \emptyset$
12. $Ie = IA + (VR)^2 IB + (VR)^2 IC + (VR)^2 ID$
13. $T \propto= Ie \times \alpha A$
14. $T\alpha = TA + \frac{(NB)}{(NA)}\frac{TBC}{\eta 1} + \frac{(ND)}{(NA)}\frac{TD}{\eta 1 \eta 2}$
15. $\frac{NB}{NA} = \frac{wB}{wA} = \frac{\alpha B}{\alpha A} = \frac{IA}{IB}$
16. $T_{OUTPUT} = T_{INPUT} \times GR \times \eta$
17. $P = \frac{\pi \times PCD}{n}$
18. $Ti + To + Th = 0$
19. $TA = TS + 2TP$
20. $\frac{Input\ speed}{Output\ speed} = \frac{Teeth\ on\ driven\ gears}{Teeth\ on\ driving\ gears}$
21. $v = \pi \times (d + t) \times N$
22. $p = Te \times v$
23. $\frac{T1}{T2} = e^{\mu\theta}$
24. $T1 = \delta \times A$
25. $Tc = m \times v^2$
26. $\frac{T1 - TC}{T2 - TC} = e^{\mu\theta cosec\alpha}$

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27. $L = \frac{\pi}{2} \times (D + d) + \frac{(D \pm d)^2}{4 \times C} + 2C$

28. $Tg = m \times g \times \sin \emptyset$

29. $v = w \times r$

30. $v = \sqrt{\mu \times g \times r}$

31. $v = \sqrt{\frac{g \times b \times r}{2 \times h}}$

32. $v = \sqrt{gr[\frac{\mu + \tan \theta}{1 - \mu \tan \theta}]}$

33. $v = \sqrt{gr[\frac{h \tan \theta + b/2}{h - b/2 \tan \theta}]}$

34. $\frac{T1}{T2} = [\frac{1 + \mu \tan \theta}{1 - \mu \tan \theta}]^n$

35. $\cos \frac{\theta}{2} = \frac{R - r}{C}$

36. $\cos \frac{\emptyset}{2} = \frac{R + r}{C}$

37. $m = w \times t \times L \times \rho$

38. $T1 = w \times n \times ft$

39. $P = Pg + P\mu$

40. $t = \frac{I \times w}{T}$

41. $P = \frac{2 \times \pi \times N \times T}{60}$

42. $T = F \times r$

43. $w = do + 3d - 1{,}5155P$

44. $do = de + +0{,}65P$

45. $w = \frac{\pi \times m}{2}(\cos^2 \theta)$

46. $h = m[1 - \frac{\pi}{4}(\sin \theta \cos \theta)]$

47. $\frac{P1}{Rho} + \frac{(v1)^2}{2} + gh1 = \frac{P2}{R} + \frac{(v2)^2}{2} + gh2$

48. $Vw(Va) = \sqrt{\frac{gx^2}{2y}}$

49. $v = C\sqrt{mi}$

50. $hf = \frac{4 \times f \times \ell \times v^2}{2 \times g \times d}$

51. $hf = \frac{f \times \ell \times O^2}{3{,}026 \times d^5}$

52. $Q = \frac{Cd \times A \times a \times \sqrt{(2gh)}}{\sqrt{(A^2 - a^2)}}$

53. $Q = Cd \times A \times \frac{\sqrt{(2gh)}}{\sqrt{(m^2 - 1)}}$

54. $V = \sqrt{(g \times R \times \cos \theta}$

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55. $Vol.bucket = \frac{m \times s}{\rho \times v}$

56. $L = 2C + \pi D$

57. $Self - weight = \frac{m1 \times g \times S^2}{8 \times h}$

58. $One\ load = \frac{m2 \times g \times S}{4 \times h}$

59. $T(acc\ load) = (T1 - T2)R$

60. $T(acc\ drum) = I \times a = mk^2 \times \frac{a}{R}$

61. $P = w \times T$

62. $w = 2\pi \times N$

63. $Ke = \frac{1}{2} I \times w^2$

64. $Ke = \frac{work\ done}{efficiency}$

65. $P = Ke \times operations/sec$

66. $(I_1 + I_2)w_3 = I_1 w_1 + I_2 w_2$

67. $\mu = \tan\theta$

68. $\eta = \frac{\tan\theta}{\tan(\theta + \emptyset)}$

69. $T = \mu \times F \times Re \times n$

70. $T = \frac{\mu \times F \times Re}{\sin\theta}$

71. $T = \mu \times n \times (Fc - S)R$

72. $Fc = m \times w^2 \times y$

73. $Fc = \frac{mv^2}{y}$

74. $Tractive\ effort = mass\ on\ driving\ wheels \times \mu \times g$

75. $Side\ thrust = Fc\cos\theta - mg\sin\theta$

76. $\mu = \frac{Fc\cos\theta - mg\sin\theta}{mg\cos\theta + Fc\sin\theta}$

77. $P_l = CmgL + mgh$

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